# Operational Research: Tutorial 4

Mathematical Programming: Modelling

Answer notebooks - https://ctsallis.github.io/operational_research/ → Lectures → Semester 1 → Week 04 - Mathematical Programming - Modelling → 4 Tutorial Answers

## Exercises

1) *C&S* (Management Consultants) have been employed to help the *Icki-Toffee* Company to develop a popular, economical but more health-conscious selection of candies to add to their scrumptious range of toffees. There has been much publicity that the Government are so worried about obesity that they plan to introduce a coding system for all food products which will inform the public about products containing foods which may be adding to the problem. There are rumours that the Government aim to strongly promote the concept of fruit as a healthy alternative to sweets, so *Icki-Toffee* are very keen to introduce fruit, in the form of fillings, soft centres etc into their range. They are determined to ensure that at least one of their selections of candies does not fall into the worst category.

    - The selection chosen must weigh at least 320g
    - Total fat content by weight must be at most 45% of the total weight of the selection.
    - Calorific value of the selection must be at most 2850 calories.

    The quantities of fat and calories in each candy are given in Table 1. The costs of each candy are given in Table 2.

    Table 2 also lists the results of an in-store survey which Icki-Toffee carried out in several of its retail outlets. The popularity number is measured on the scale where 10 is very popular and 0 not popular.

**a)** **Based on your model from last week, impose a constraint which limits the weight to 1.5 times the minimum weight of 320g.**

Introduce the following constraint into last week's model

$$\sum_i W_i x_i \leq 1.5 \cdot 320$$

**b)** **Combine the cost and popularity to form an objective function. Justify your choice based on your own opinion, i.e. do you think cost and popularity are equally important?** *

| Candy - $x_i$ | Fat (g) - $F_i$ | Calories – $K_i$ | Weight (g) - $W_i$ |
|---|---|---|---|
| **Buttery butterscotch** | 20 | 100 | 24 |
| **Caramel softie** | 16 | 100 | 22 |
| **Orange dream** | 10 | 80 | 15 |
| **Coconut Ice** | 7 | 50 | 12 |
| **Lemon sorbet** | 6 | 60 | 10 |
| **Lime sherbet** | 5 | 40 | 10 |
| **Strawberry surprise** | 13 | 80 | 14 |
| **Raspberry mousse** | 15 | 90 | 17 |
| **Ginger toffee** | 18 | 80 | 20 |
| **Rose-water posy** | - | 145 | 8 |
| **Walnut whirl** | 19 | 120 | 20 |
| **Grape bunch** | - | 120 | 15 |

*Table 1: Table to show content information about each candy*

| Candy - $x_i$ | Cost - $C_i$ | Popularity - $P_i$ |
|---|---|---|
| **Buttery butterscotch** | 12 | 10 |
| **Caramel softie** | 12 | 10 |
| **Orange dream** | 12 | 8 |
| **Coconut Ice** | 15 | 9 |
| **Lemon sorbet** | 13 | 5 |
| **Lime sherbet** | 13 | 6 |
| **Strawberry surprise** | 16 | 7 |
| **Raspberry mousse** | 16 | 8 |
| **Ginger toffee** | 15 | 10 |
| **Rose-water posy** | 10 | 3 |
| **Walnut whirl** | 18 | 8 |
| **Grape bunch** | 20 | 4 |

*Table 2: Cost and popularity data.*

Look at the provided answer notebooks on the course site (4 Tutorial Answers). You have many possible choices. I have implemented a weighted objective, a normalised weighted objective and a popularity per cost ratio objective. These are all from the lecture slides!

With the weight cap, the answer notebooks give these selections:

| Objective | Selection | Weight (g) | Cost | Popularity |
|---|---|---|---|---|
| **Popularity per cost, $\Sigma(P_i/C_i)x_i$** | 8 Coconut Ice, 32 Lime sherbet, 8 Rose-water posy | 480 | 616 | 288 |
| **Weighted, $\alpha = 0$** | 30 Coconut Ice, 5 Rose-water posy, 5 Grape bunch † | 475 | 600 | 305 |
| **Normalised weighted, $\alpha = 0.5$** | 24 Coconut Ice, 11 Rose-water posy | 376 | 470 | 249 |
| **Normalised weighted, $\alpha = 0.59$** | 9 Caramel softie, 10 Rose-water posy, 3 Grape bunch | 323 | 268 | 132 |

† Other selections also reach popularity 305. With the normalised weighted objective, any α from about 0.59 upwards gives the minimum-cost selection (cost 268) and any α below 1/6 ≈ 0.167 gives the maximum-popularity selection (305). With the plain weighted objective the costs dominate: the minimum-cost selection appears from α ≈ 0.42 and the maximum-popularity one only below α ≈ 0.09. The popularity-per-cost objective adds up each candy's own ratio, so the selection's total popularity ÷ total cost (288/616 = 0.47) is not the highest possible.

2) A company operates in two factories, A and B. Each factory makes two products, ***standard*** and ***deluxe***. A unit of ***standard*** gives a profit contribution of £10, while a unit of ***deluxe*** gives a profit contribution of £15.
Each factory uses two processes, grinding and polishing, for producing its products. Factory A has a grinding capacity of 80 hours per week and polishing capacity of 60 hours per week. Factory B has a grinding capacity of 60 hours per week and polishing capacity of 75 hours per week.
The grinding and polishing times in hours for a unit of each type of product in each factory are given in the table below.

| | Factory A | | Factory B | |
|---|---|---|---|---|
| | Standard | Deluxe | Standard | Deluxe |
| Grinding | 4 | 2 | 5 | 3 |

| | | | | |
|---|---|---|---|---|
| Polishing | 2 | 5 | 5 | 6 |

This might be the result of older machines at Factory B.

In addition, each unit of product (***standard*** *or* ***deluxe***) uses 4kg of raw material. The company has 120kg of raw material available per week. The company currently splits the 120kg between the factories. Factory A is allocated 75kg of raw material a week and factory B is allocated 45kg per week.

**a) Build a separate Linear Program (LP) for each factory to maximise its profit.**
Note whether you use integer solutions or real is dependent on your interpretation. I have chosen to use integer, but real is fine.

$$s - standard$$
$$d - deluxe$$

**Factory A Model**
maximise $10s + 15d$
subject to:

$$4s + 2d \leq 80$$
$$2s + 5d \leq 60$$
$$4s + 4d \leq 75$$

and:

$$s, d \geq 0$$
$$s, d \in \mathbb{Z}$$

**Factory B Model**
maximise $10s + 15d$
subject to:

$$5s + 3d \leq 60$$
$$5s + 6d \leq 75$$
$$4s + 4d \leq 45$$

and:

$$s, d \geq 0$$
$$s, d \in \mathbb{Z}$$

**i. How much of standard or deluxe should each factory make? What is the profit for each factory and therefore the combined profit?**

**Factory A**

$$s = 10$$
$$d = 8$$

i.e. make 10 standard and 8 deluxe.
Profit is then $10 \cdot 10 + 15 \cdot 8 = 220$

**Factory B**

$$s = 0$$
$$d = 11$$

i.e. make 0 standard and 11 deluxe.
Profit is then $10 \cdot 0 + 15 \cdot 11 = 165$

Sensitivity: Internal

ii. **Is there any grinding or polishing capacity left at each factory? (i.e. the number of hours not used. E.g. Factory A might use only 65 hours of the 80 hours per week for grinding, resulting in 15 unused hours.)**

**Factory A Slack (unused hours)**

Calculate the slack values by inputting the solution $s = 10$ and $d = 8$ into the constraints

$4 \cdot 10 + 2 \cdot 8 = 56$ (grind hours used)
$2 \cdot 10 + 5 \cdot 8 = 60$ (polish hours used)
$4 \cdot 10 + 4 \cdot 8 = 72$ (raw materials used)

Therefore the slack (unused hours) for grind and polish is:
$80 - 56 = 24$ (grind slack)
$60 - 60 = 0$ (polish slack)

**Factory B Slack (unused hours)**

Calculate the slack values by inputting the solution $s = 0$ and $d = 11$ into the constraints

$5 \cdot 0 + 3 \cdot 11 = 33$ (grind hours used)
$5 \cdot 0 + 6 \cdot 11 = 66$ (polish hours used)
$4 \cdot 0 + 4 \cdot 11 = 44$ (raw materials used)

Therefore the slack (unused hours) for grind and polish is:
$60 - 33 = 27$ (grind slack)
$75 - 66 = 9$ (polish slack)

b) **Given that the raw material is no longer allocated manually, combine the factories into one large LP for the company to maximise the profit.**

maximise $10s_1 + 15d_1 + 10s_2 + 15d_2$

subject to:

$$\begin{aligned}
4s_1 + 2d_1 &\leq 80 \\
2s_1 + 5d_1 &\leq 60 \\
5s_2 + 3d_2 &\leq 60 \\
5s_2 + 6d_2 &\leq 75 \\
4s_1 + 4d_1 + 4s_2 + 4d_2 &\leq 120
\end{aligned}$$

and:

$s_1, s_2, d_1, d_2 \geq 0$
$s_1, s_2, d_1, d_2 \in \mathbb{Z}$

i. **What is the total profit? Is this better or worse than the profit given by the separate models?**

$s_1 = 10$
$d_1 = 8$
$s_2 = 0$
$d_2 = 12$

Profit is then given by $10 \cdot 10 + 15 \cdot 8 + 10 \cdot 0 + 15 \cdot 12 = 400$

Better profit as 400 is greater than 220+165 = 385

ii. **How much raw material was used by each of the factories? What does this tell you about the separate models?**

Sensitivity: Internal

**Factory A Raw Material Used**

$$4 \cdot 10 + 4 \cdot 8 = 72$$

**Factory B Raw Material Used**

$$4 \cdot 0 + 4 \cdot 12 = 48$$

Thus, we have an optimal split of 72:48 for the raw material, with the separate models we have a sub-optimal split of 75:45 which meant that in the end we made only £385 profit rather than £400

Sensitivity: Internal